

8(a) gravitational force provides (just enough) centripetal acceleration of 8.825 m s^{-2} on both ISS and astronaut to keep them in circular orbit around Earth

OR

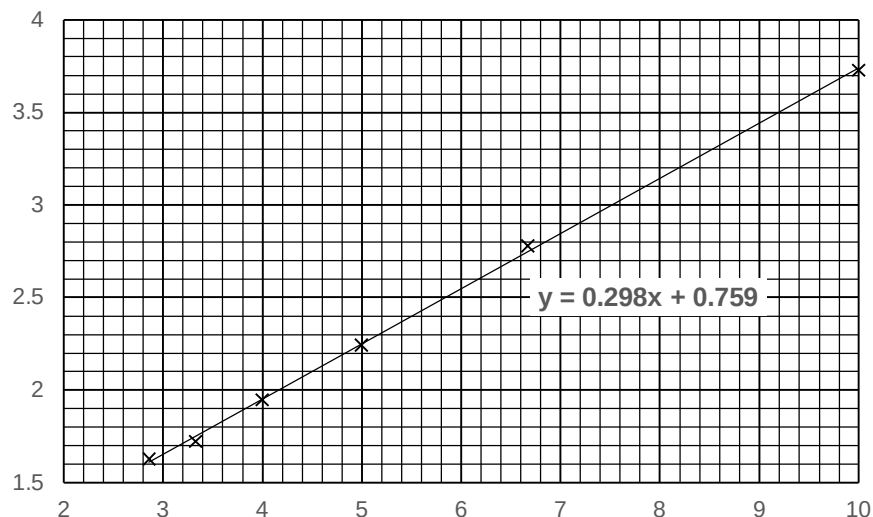
both ISS and astronaut experience 8.825 ms^{-2} of acceleration directed to the centre of the earth due to gravity

no contact force by ISS on astronaut.

M1

A1

8(b)(i)



best fit straight line with line thickness not comparable to half sq

$$p = 0.298 \text{ kg s}^{-2}$$

$$q = 0.759 \text{ s}^{-2}$$

B1

B1

B1

8(b)(ii)

$$\text{For } \frac{1}{T^2} = \frac{p}{M} + q \rightarrow T = \sqrt{\left(\frac{p}{M} + q\right)^{-1}}$$

$$\text{For } M = 0.5 \text{ kg, } T = \sqrt{\left(\frac{0.298}{0.5} + 0.759\right)^{-1}} = 0.859 \text{ s}$$

$$\partial T = \frac{2(0.298)(0.858)^3}{0.5} \times (10\%) = 0.08 \text{ s (1 s.f.)}$$

C1

A1

8(b)(iii)

Yes. Period has more precision than what is required of variation

from table of values (Fig. 8.3), the period is to 4 decimal points while the variation which is to 2 d.p.

B1

B1